# 2LA 2021 Solutions to Test 2

## Section A

1. (a) $(\frac{1}{2}, 12)$
   (b) $(1, 0)$
   (c) $(x_1^{-1}, -x_2)$

2. (a) $S_3$
   (b) $S_1$
   (c) Impossible.
   (d) $S_2$

3. (a) Rank of $A$ =3, Nullity of $A$=2.
   (b) Nullity of $A^T$=1.
   (c) Basis for $NS(A) = \left\{ \begin{pmatrix} -1 \\ 3 \\ 2 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}$
   (d) Basis for $CS(A) = \left\{ \begin{pmatrix} 1 \\ 3 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \\ -3 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 \\ 13 \\ 3 \\ 8 \end{pmatrix} \right\}$
   (e) Basis for $RS(A) = \{(1, -1, 2, 3, 4), (0, 2, -3, 0, 1), (0, 0, 0, 0, 1)\}$
   (f) $CS(A) = \left\{ \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \in \mathbb{R}^4 : 4a - b - c = 0 \right\}$.

## Section B

4. (a) It is possible: for $\mathbf{v}$ take any vector in $\mathbb{R}^4$ not in $CS(A)$, such as for example $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$.

   Then (by a theorem) $B \cup \{\boldsymbol{v}\}$ will be a linearly independent set in $\mathbb{R}^4$ with 4 elements. Since $\mathbb{R}^4$ has dimension 4, $B \cup \{\boldsymbol{v}\}$ will be a basis for $\mathbb{R}^4$.

   (b) No, it is not possible. There are only 3 columns in $U$ with pivots in them. If we remove any column in $A$ corresponding to a column in $U$ without a pivot in it, the resulting matrix $B$ will have rank $= 3 < 4$, and therefore $B$ will not be invertible (since a $4 \times 4$ matrix is invertible iff it has rank $= 4$).

5. (a) $S$ is closed under addition. Let $p(x), q(x) \in S$, then $p(0) = 0, p(1) \geq 1, q(0) = 0, q(1) \geq 1$ and therefore $(p+q)(1) = p(1) + q(1) \geq 1 + 1 = 2 \geq 1$, $\quad (p+q)(0) = p(0) + q(0) = 0 + 0 = 0$, and so $p(x) + q(x) \in S$.

(b) $S$ is not a subspace. To show this, it will be enough to show that $S$ is not closed under scalar multiplication. Let $p(x) \in S$ and $\lambda = -1$. Then $p(1) \geq 1$ and $(\lambda p)(1) = (-1)p(1) \leq (-1) \times 1 = -1$, and so $\lambda p(x) \notin S$. This shows that $S$ is not closed under scalar multiplication.

(c) Let $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$ and suppose

$$\lambda_1(x^3 - 3x^2 - 6x) + \lambda_2(2x^3 - 3x) + \lambda_3(4x^2 + 8x) = 0.$$

Then

$$(\lambda_1 + 2\lambda_2)x^3 + (-3\lambda_1 + 4\lambda_3)x^2 + (-6\lambda_1 - 3\lambda_2 + 8\lambda_3)x = 0x^3 + 0x^2 + 0x.$$

Comparing coefficients of powers of $x$ gives the system of equations

$$\begin{aligned} \lambda_1 + 2\lambda_2 &= 0 \\ -3\lambda_1 + 4\lambda_3 &= 0 \\ -6\lambda_1 - 3\lambda_2 + 8\lambda_3 &= 0 \end{aligned}$$

The coefficient matrix $\begin{pmatrix} 1 & 2 & 0 \\ -3 & 0 & 4 \\ -6 & -3 & 8 \end{pmatrix}$ Gauss reduces to $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 2 \\ 0 & 0 & 2 \end{pmatrix}$.
From this it follows that $\lambda_1 = \lambda_2 = \lambda_3 = 0$, and hence $R$ is linearly independent.

(d) Since $R$ is a linearly independent set, it is a basis for span$(R)$. From this it follows that dim(span$(R) = 3$.

6. (a) Let $f_1, f_2 \in V$. Then for every $x \in \mathbb{R}$,

$$\begin{aligned} T(f_1)(x) + T(f_2)(x) &= (f_1(0) + f_1'(0)x) + (f_2(0) + f_2'(0)x) \\ &= (f_1(0) + f_2(0)) + (f_1'(0) + f_2'(0))x \\ &= (f_1 + f_2)(0) + (f_1 + f_2)'(0)x \\ &= T(f_1 + f_2)(x), \end{aligned}$$

so $T(f_1) + T(f_2) = T(f_1 + f_2)$.
If $\lambda$ is a scalar, then for every $x \in \mathbb{R}$,

$$\begin{aligned} \lambda T(f_1)(x) &= \lambda(f_1(0) + f_1'(0)x) \\ &= (\lambda f_1(0) + \lambda f_1'(0)x \\ &= ((\lambda f_1)(0) + (\lambda f_1)'(0)x \\ &= T(\lambda f_1)(x), \end{aligned}$$

so $\lambda T(f_1) = T(\lambda f_1)$.

(b) The range of $T$ is $P_1$. It is clear that for every $f \in V$, $T(f) \in P_1$, so ran$(T) \subseteq P_1$. Conversely, if $p(x) = ax + b \in P_1$, then $p(0) = b, p'(0) = a$, so $p = T(p) \in$ ran$(T)$ and therefore $P_1 \subseteq$ ran$(T)$. Then dim(ran$(T)) = $ dim$(P_1) = 2$.

(c) $f(x) = x^2$ will do (since $f(0) = 0, f'(0) = 0$).

(d) For $n \geq 2$, let $f_n(x) = x^n$. Then $f_n \in \ker(T)$ for every $n \geq 2$, and $\{f_n : n \geq 2\}$ is an infinite linearly independent set in $\ker(T)$. Hence $\ker(T)$ is infinite dimensional. Note that simply showing that $\ker(T)$ has infinitely many elements is not enough to show that it is infinite-dimensional; even a one-dimensional space has infinitely many elements.

7. (a) Let $S$ be a linearly independent set of a vector space $V$. If $\mathbf{v} \in V$ and $\mathbf{v} \notin \mathrm{span}(S)$, then $S \cup \{\mathbf{v}\}$ is linearly independent.

   (b) It is assumed that $S \cup \{\mathbf{v}\}$ is linearly dependent.

   (c) If we had $a_{n+1} = 0$, then we would have

$$a_1\mathbf{u}_1 + a_2\mathbf{u}_2 + \cdots + a_n\mathbf{u}_n = \mathbf{0},$$

   with not all of $a_1, a_2, \ldots, a_n$ equal to zero (since we assumed that not all of $a_1, a_2, \ldots, a_{n+1}$ are zero). This would imply that $S$ is linearly dependent.

   (d) The equation (3) shows that $\mathbf{v} \in \mathrm{span}(S)$, contradicting the assumption that $\mathbf{v} \notin \mathrm{span}(S)$.

   (e) If $S$ is a linearly independent set with $n$ elements in an $n$-dimensional vector space $V$, then $S$ is a basis for $V$.

   (f) $S \cup \{\mathbf{v}\}$ consists of $n+1$ vectors and we have proved here that it is linearly independent. But any set with $n+1$ elements in an $n$-dimensional vector space is linearly dependent.

   (g) The assumption that $S$ does not span $V$ has lead to a contradiction, hence it spans $V$ and is linearly independent, and is therefore a basis.

8. (a) False. As a counter-example, let

$$C = \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

   Then $CS(C) = CS(D) = \left\{ \begin{pmatrix} a \\ 0 \end{pmatrix} : a \in \mathbb{R} \right\}$, but $NS(C) = \left\{ \lambda \begin{pmatrix} -2 \\ 1 \end{pmatrix} : \lambda \in \mathbb{R} \right\}$, $NS(D) = \left\{ \lambda \begin{pmatrix} -1 \\ 1 \end{pmatrix} : \lambda \in \mathbb{R} \right\}$, and so $NS(C) \neq NS(D)$.

   (b) True. Suppose $\dim(W) = \dim(V) = n$. Then $W$ has a basis $B$ with $n$ elements. Since $B$ is a linearly independent set with $n = \dim(V)$ elements, it must span $V$. But then $W = \mathrm{span}(B) = V$.

   (c) False. As a counter example, Let $V = P(x)$, the vector space of all polynomials in the variable $x$, and let $W = \mathrm{span}(\{x^{2n} : n = 0, 1, 2, \ldots\})$. Then $W \neq V$, and $W$ is infinite-dimensional, since it contains the infinite linearly independent set $\{x^{2n} : n = 0, 1, 2, \ldots\}$.